

Mohammed Hisham Samir Abdel-Aziz

Data Science & AI Student | Data Engineer | ML Enthusiast | Junior Teaching Assistant

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Junior Data Scientist, Data Engineer, Data Analyst, and AI Engineer enrolled at Zewail City of Science, Technology and Innovation. Selected as Junior Teaching Assistant based on academic excellence. Proficient in Python, data analysis, data visualization (Power BI, Tableau), automated web scraping, ETL pipeline development, and object-oriented system design. Passionate about building end-to-end data solutions and extracting actionable insights from complex datasets.

EDUCATION

B.Sc. Computational Science and Artificial Intelligence — Data Science & AI Track | Zewail City of Science, Technology and Innovation *Expected 2029*

- Semester 1 GPA: 3.33 | Semester 2 GPA: 3.60 | Cumulative GPA (CGPA): 3.45
- Relevant Coursework: Calculus (A), Logic (A), ETL (A), Python Programming (A-), OOP (A-), Linear Algebra (B+), KRR (B+), Computer Networks (CCNA)

ACADEMIC EXPERIENCE

Junior Teaching Assistant (JTA) | Zewail City of Science, Technology and Innovation *2026 – Present*

- Selected based on outstanding academic performance to support peers across multiple technical courses.
- Facilitated student understanding of Digital Logic circuits, Computer Architecture fundamentals, and Applied Calculus for Computer Science.
- Provided one-on-one tutoring and group sessions to reinforce theoretical concepts and practical problem-solving skills.

INTERNSHIP EXPERIENCE

Machine Learning Intern | FlyRankAi

2026

- Completed a hands-on machine learning internship focused on supervised learning algorithms, model evaluation, and real-world dataset analysis.
- Gained practical exposure to the full ML workflow: data preprocessing, feature engineering, model training, and performance evaluation.

TECHNICAL PROJECTS

Object-Oriented E-Commerce Platform — C++ — Academic | CSAI 151 | Spring 2026 [\[GitHub\]](#)

- Designed and implemented a full-featured console-based e-commerce system in C++ simulating real-world platforms (Amazon, eBay).
- Architected 15+ classes across UI, Core, User, and Functionality layers applying OOP principles: inheritance, encapsulation, and separation of concerns.
- Built 6 core modules covering user authentication, product search, cart management, discount handling, and order processing.
- Implemented a FileHandler module for persistent data storage (products, users, order history) supporting 100+ concurrent product records via text files.
- Supported dual roles — Customer and Admin — with distinct permission sets for browsing/purchasing and inventory management.

Student Management System — Advanced OOP Design — Python — Academic Project [\[GitHub\]](#)

- Engineered a fully object-oriented Python application using inheritance, polymorphism, encapsulation, and abstraction across 10+ classes.
- Designed a multi-layer class hierarchy with abstract base classes to manage Students, Instructors, Courses, and Lessons across distinct user roles.
- Applied composition and aggregation patterns to model entity relationships, ensuring high code reusability and clean separation of concerns.

Tech Opportunities Hub — Job & Internship Aggregator — *Personal Project*

- Built a Python application aggregating 200+ tech internship, scholarship, and career opportunities from 8+ sources using BeautifulSoup and Selenium.
- Architected the system using OOP principles — designed modular classes for scraping, parsing, filtering, and data storage with clean separation of concerns.
- Developed a structured ETL pipeline to collect, clean, and categorize listings, reducing manual search time by 80% for target users.

Smartphone Price Intelligence — Multi-Source ETL Pipeline & Market Clustering — *Personal Project*

[\[GitHub\]](#)

- Engineered an end-to-end ETL pipeline processing 5,000+ smartphone listings from eBay across 3 parallel scraping methods: BeautifulSoup (BS4), Selenium, and SerpApi Google Shopping.
- Unified data from 3 sources into a single structured dataset, resolving schema conflicts, normalising price formats, and achieving 95%+ data consistency.
- Built NetworkX knowledge graphs per data source to visualise product-to-source relationships and expose coverage gaps across scraping methods.
- Applied K-Means clustering (k=3) to segment smartphones into budget, mid-range, and premium market tiers, visualised in an interactive 3D Plotly chart.

Advanced Calculator — Python — *Personal Project* [\[GitHub\]](#)

- Built a multi-functional calculator supporting 20+ arithmetic, scientific, and expression-based operations.
- Implemented clean input parsing and robust error handling covering 15+ edge cases and invalid expression scenarios.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C++

OOP: Python — Classes, Inheritance, Polymorphism, Encapsulation, Abstraction; C++ — OOP, Pointers, Templates

Data Structures: Arrays, Linked Lists, Stacks, Queues, Trees, Hash Tables, Graphs (Python & C++)

Data Engineering: ETL Pipelines, Data Cleaning, Schema Normalisation, Pipeline Automation

Databases: SQL, SQLite, PostgreSQL

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, mplot3d, Seaborn, NetworkX, Power BI, Tableau, Microsoft Excel

Web Scraping & Automation: BeautifulSoup (BS4), Selenium, SerpApi

Computer Vision: OpenCV, dlib

Machine Learning: Random Forest (in training)

Mathematics: Calculus, Linear Algebra, Statistics, Probability

Networking: Cisco CCNA (In Progress), TCP/IP, Network Fundamentals

DevOps & Tools: Git, GitHub, Docker, Google Colab

CERTIFICATIONS & TRAINING

Machine Learning Internship | FlyRankAi

2026

- Hands-on internship covering supervised learning algorithms, model evaluation, and real-world ML workflows.

AI and Data Science Diploma | INSTANT Institute

Currently Enrolled, 2026

6-track program covering Python, OOP, Data Structures, Mathematics, Web Scraping, ETL/ELT, SQL, Power BI, Tableau, Machine Learning, Deep Learning, NLP, and Computer Vision.

Data Science & AI Professional Course | INSTANT Institute

Currently Enrolled, 2026